

Smoking and Mortality Risk (NHANES 2013–2014)

December 13, 2025

1 Data

This section details the source datasets, how they are obtained, and the basic shape and missingness patterns. DEMO and SMQ tables are joined on SEQN per NHANES cycle; mortality is linked via public-use files.

Data source: Cleaned matrices (label = dead) from clean-data/

NHANES cycles: H (2013–2014)

Observations (n): 6098

Variables (k): 8

Total missing cells: 0 (0.00%)

2 Cleaned Data

We construct X and y from the raw join: pick a single label (dead preferred) and drop rows only if the label is missing. Features are imputed (numeric: median + missing flags; categorical: Unknown level); age and PIR are standardized; we add age_sq and log_income. Categorical factors (smoker, sex) are used directly.

3 Feature/Label Dictionaries and Metrics

We document each feature and the label, then summarize numeric distributions and categorical compositions to contextualize model inputs.

Class Imbalance and Class Weights

Let $\pi = P(y = 1)$ denote the prevalence of deaths. When y is imbalanced (π small), unweighted logistic loss can prioritize the majority class. A common remedy is to use inverse-prevalence weights: $w_1 = 1/\pi$ for positives and $w_0 = 1/(1 - \pi)$ for negatives, often normalized so the average weight equals one. Weighted log-likelihood becomes $L(\beta) = -\sum_i w_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$, with $p_i = \text{logit}^{-1}(X_i\beta)$. **Prevalence and Suggested Class Weights**

quantity	value
n_total	6098.00
n_positive (y=1)	466.00
n_negative (y=0)	5632.00
prevalence (y=1)	0.08
w1 = 1/prevalence	13.09
w0 = 1/(1-prevalence)	1.08
w1_normalized	6.54
w0_normalized	0.54

4 Transformations

This section explains the transformation logic and motivations: label-only dropping preserves sample size; standardized numerics help comparability; categorical factors (smoker, sex) are used directly; smoker categories follow CDC coding of SMQ020/040.

5 Models

We estimate two logistic models on the cleaned matrices: (1) a baseline weighted logit with HC1-robust standard errors, and (2) a weighted regularized logit selected by cross-validation. Reference (omitted) smoking category: Never.

Model 1: Logit (Weighted, HC1 robust)

Math: $y = \beta_0 + \beta_1 \cdot 1\{smoker = Former\} + \beta_2 \cdot 1\{smoker = Current\} + \beta_3 age + \beta_4 1\{sex = male\} + \beta_5 income + \varepsilon$. (Reference: Never)

	title	
	(Intercept)	-0.889***
		(0.235)
	smokerCurrent	1.193***
		(0.210)
	smokerFormer	0.605**
		(0.278)
	age	1.422***
		(0.147)
	age_sq	0.400***
		(0.079)
Coefficients (robust SEs):	sexmale	0.176
		(0.127)
	log_income	-0.922***
		(0.138)
	smokerCurrent $\times$ age	-0.352*
		(0.199)
	smokerFormer $\times$ age	-0.033
		(0.212)
	Num.Obs.	6098
	RMSE	0.39
	Std.Errors	Custom
* p < 0.1, ** p < 0.05, *** p < 0.01		

Classification metrics (in-sample)

Metric	Value	Classification metrics (out-of-sample)
Accuracy	0.78	
LogLoss	0.46	
Brier	0.15	
ROC_AUC	0.87	
PR_AUC	0.40	

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Model 2: Regularized Logit (CV, Weighted)

Math: $\hat{\beta} = \arg \min_{\beta} \frac{1}{2n} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1$, with λ chosen by cross-validation.

Nonzero coefficients:	lambda	term	estimate	Classification metrics (in-sample)
	lambda.min	(Intercept)	−0.86	
	lambda.min	smokerCurrent	1.15	
	lambda.min	smokerFormer	0.56	
	lambda.min	age	1.39	
	lambda.min	age_sq	0.40	
	lambda.min	sexmale	0.17	
	lambda.min	log_income	−0.91	
	lambda.min	smokerCurrent:age	−0.31	
	lambda.1se	(Intercept)	−0.59	
	lambda.1se	smokerCurrent	0.62	
	lambda.1se	smokerFormer	0.14	
	lambda.1se	age	1.14	
	lambda.1se	age_sq	0.32	
	lambda.1se	log_income	−0.64	
	lambda.1se	smokerFormer:age	0.14	

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Accuracy	0.79
LogLoss	0.46
Brier	0.15
ROC_AUC	0.88
PR_AUC	0.38

6 Diagnostics

We start with multicollinearity (VIF), then show per-model diagnostic plots.

Variance Inflation Factors (VIF).

	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$
smoker	1.433874	2	1.094278
age	1.691699	1	1.300653
age_sq	1.142969	1	1.069097
sex	1.036485	1	1.018079
log_income	1.076823	1	1.037701
smoker:age	2.181395	2	1.215300

Model 1: Logit

We estimate a weighted logistic regression; residual plots use deviance residuals. Predicted probabilities are also displayed as a 1–5 risk index.

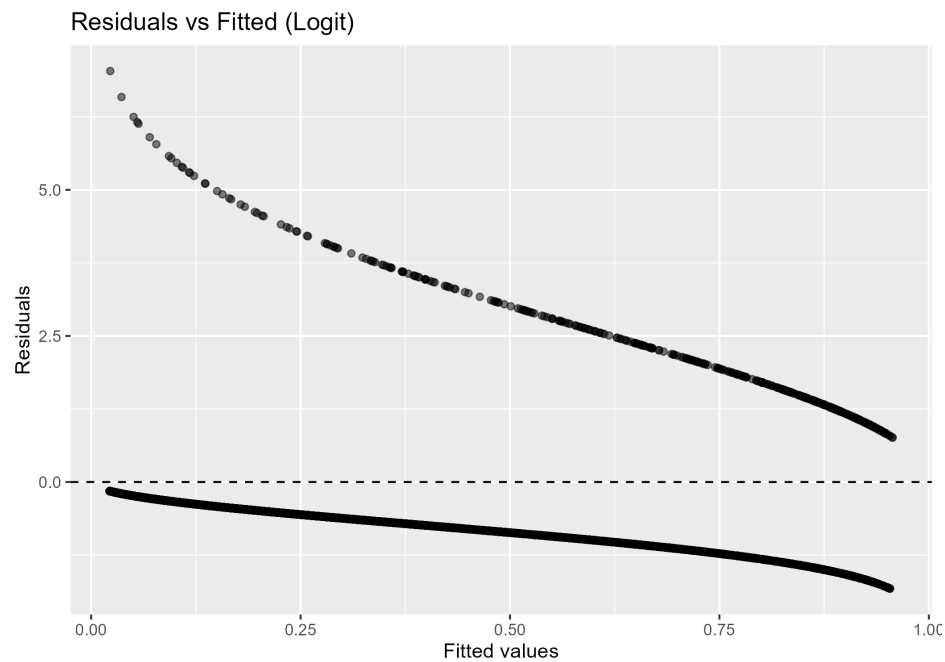


Figure 1: Logit: Residuals vs Fitted

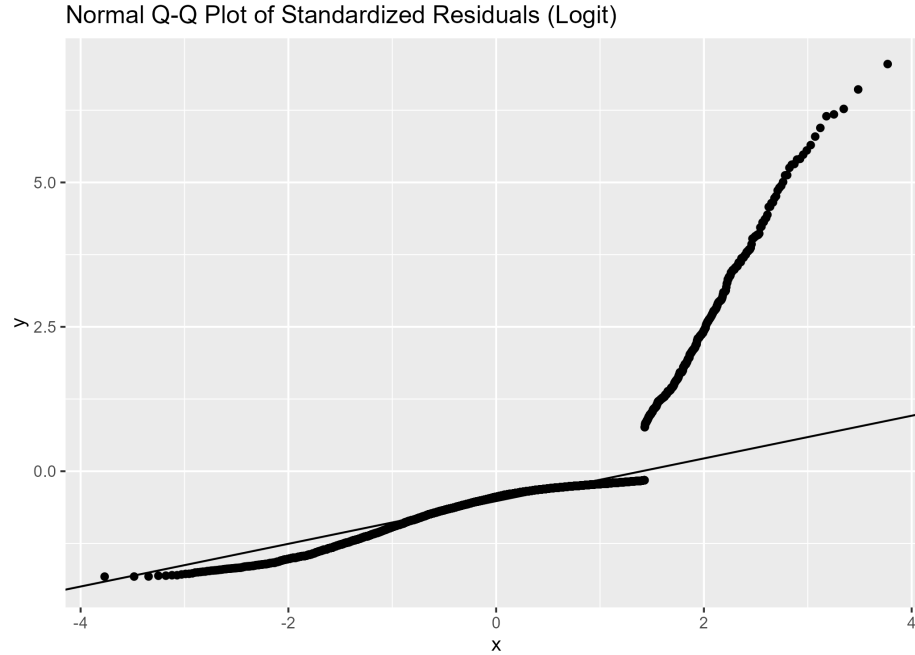


Figure 2: Logit: Normal Q-Q Plot

Model 2: Regularized Logit (CV)

We fit a logistic regression with an L_1 penalty and 10-fold cross-validation, using the same class weights as Model 1. Residuals/QQ use predictions at λ_{min} .

7 Coefficient Comparison

Logit vs Regularized Logit (lambda.min and lambda.1se)

term	base_estimate	lasso_min	lasso_1se
(Intercept)	-0.89	-0.86	-0.59
age	1.42	1.39	1.14
age_sq	0.40	0.40	0.32
log_income	-0.92	-0.91	-0.64
sexmale	0.18	0.17	
smokerCurrent	1.19	1.15	0.62
smokerCurrent:age	-0.35	-0.31	
smokerFormer	0.60	0.56	0.14
smokerFormer:age	-0.03		0.14

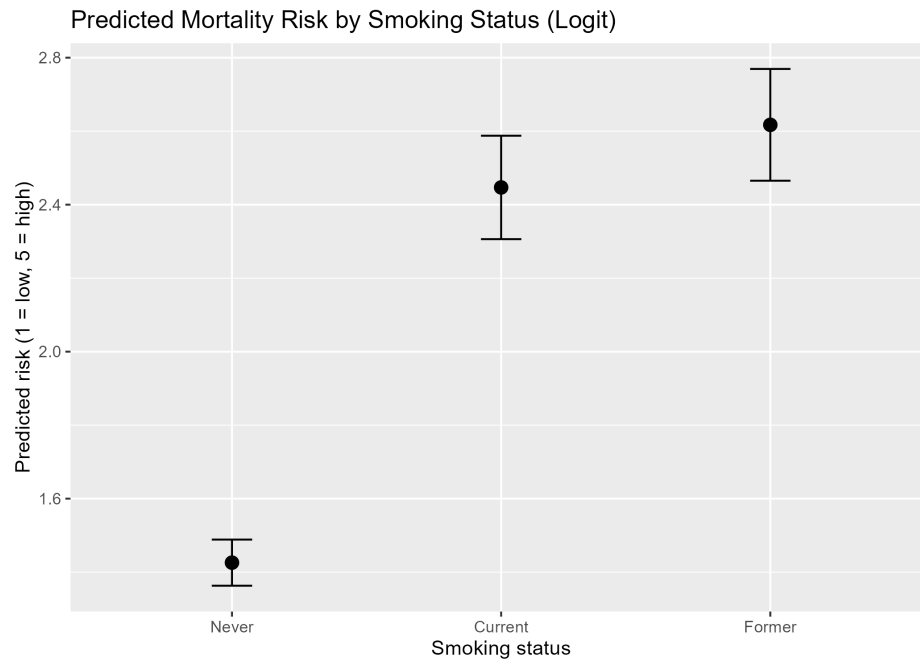


Figure 3: Logit: Predicted Risk by Smoking Status (scaled 1–5)

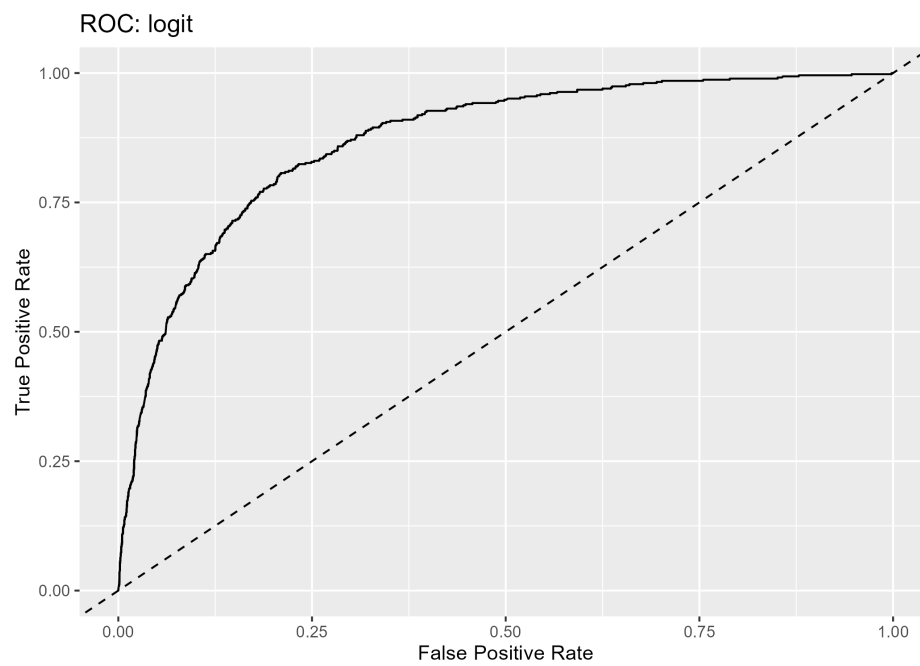


Figure 4: Logit: ROC Curve

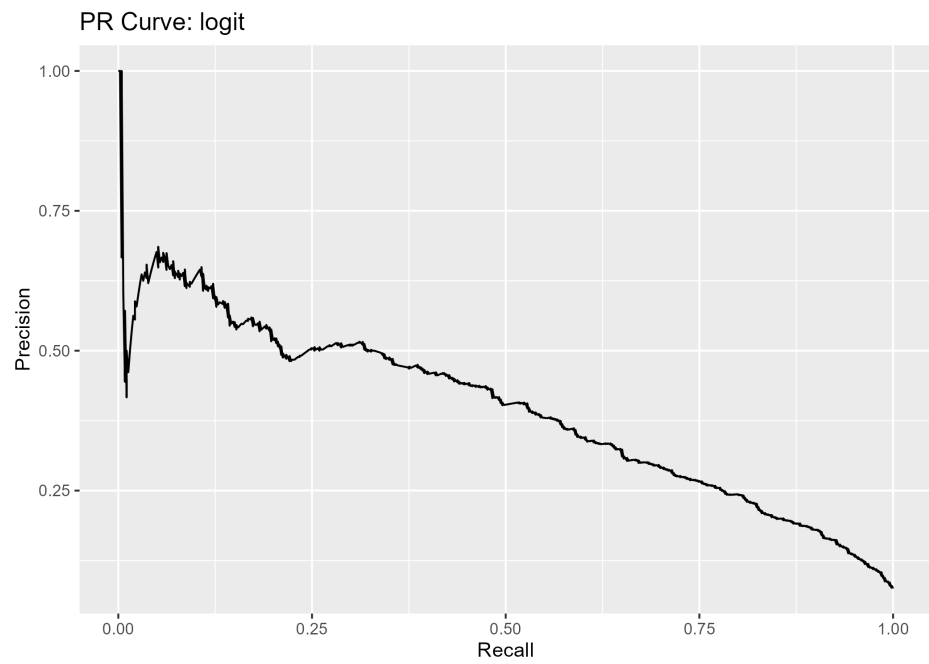


Figure 5: Logit: Precision–Recall Curve

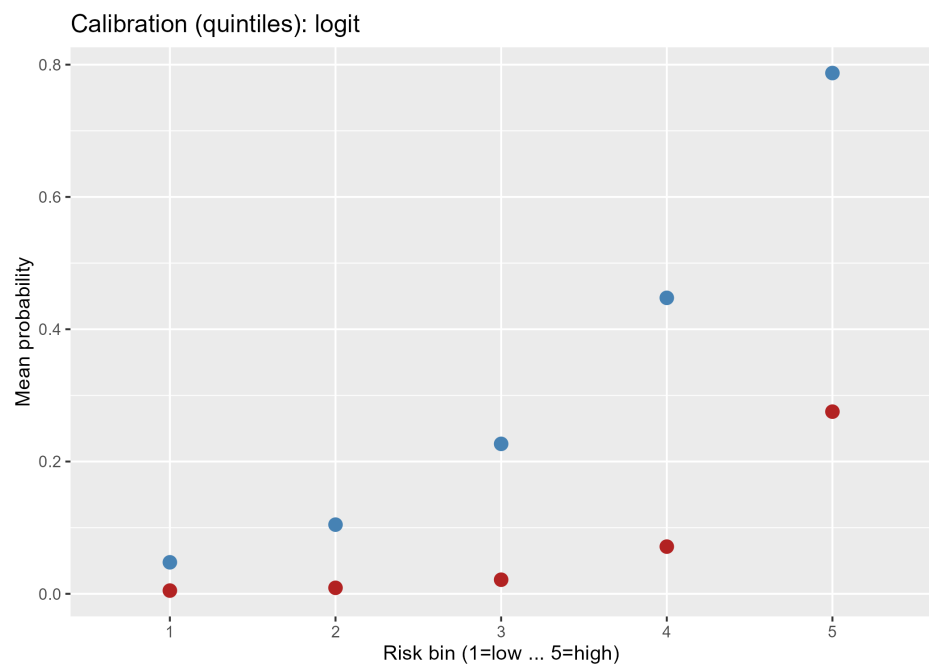


Figure 6: Logit: Calibration by Risk Quintile

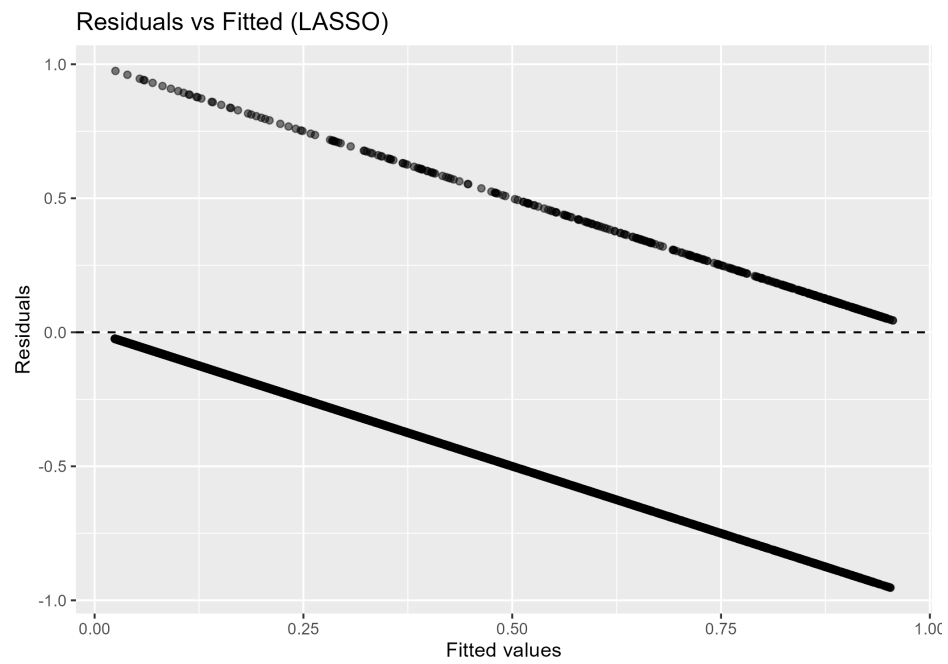


Figure 7: Regularized Logit: Residuals vs Fitted (lambda.min)

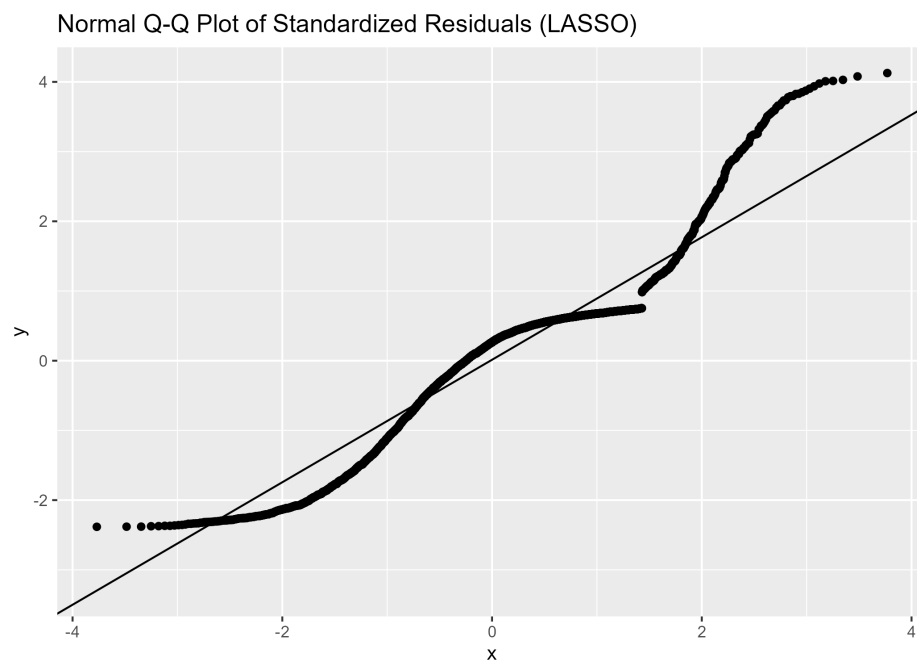


Figure 8: Regularized Logit: Normal Q-Q Plot (lambda.min)

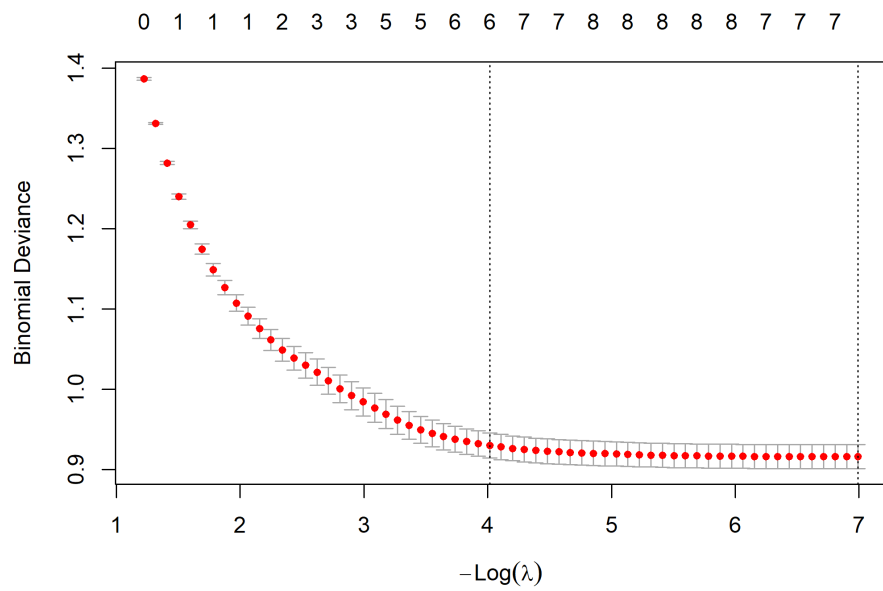


Figure 9: Regularized Logit: CV Error vs Lambda (glmnet)

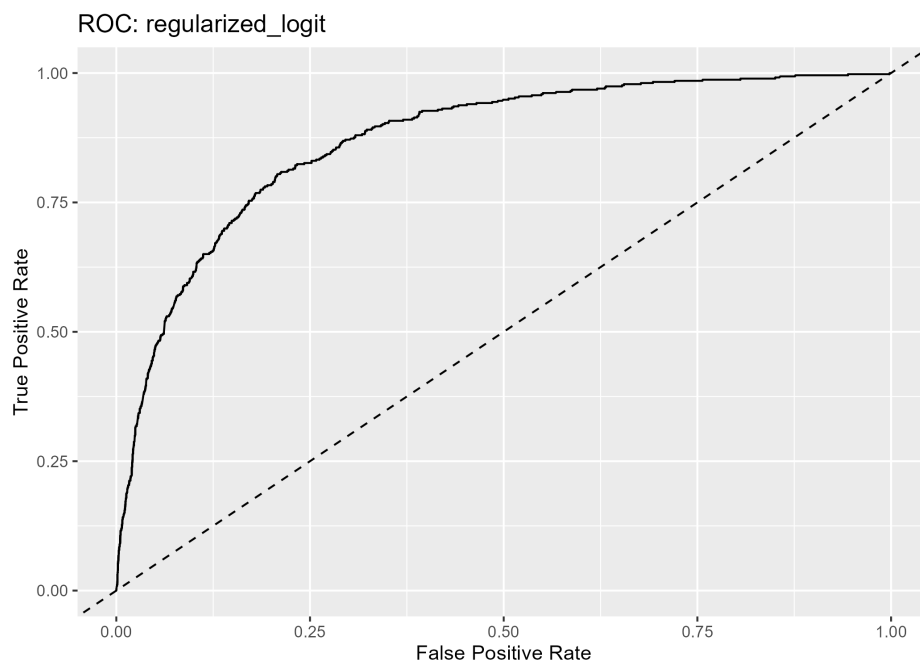


Figure 10: Regularized Logit: ROC Curve

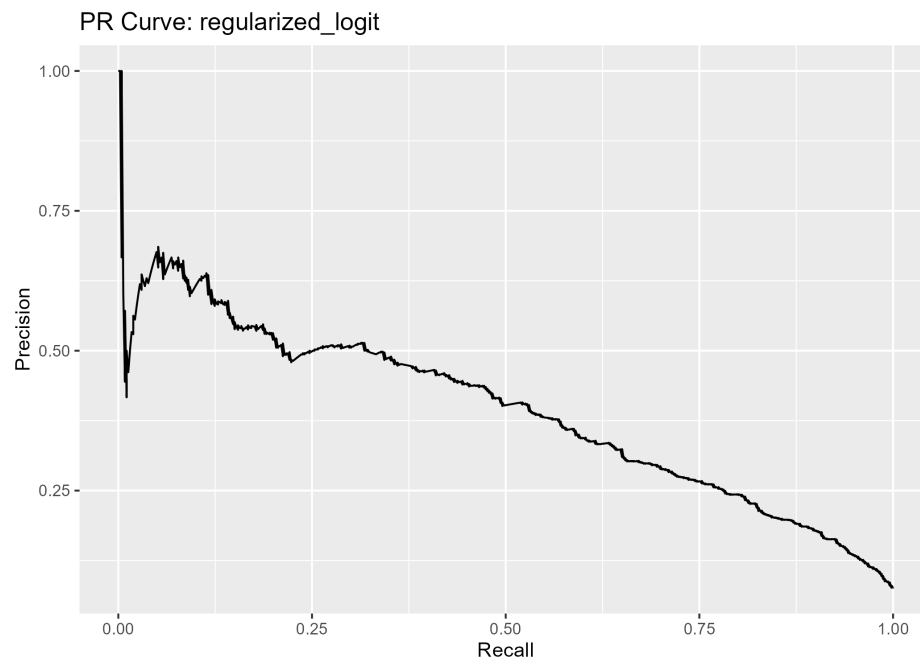


Figure 11: Regularized Logit: Precision-Recall Curve

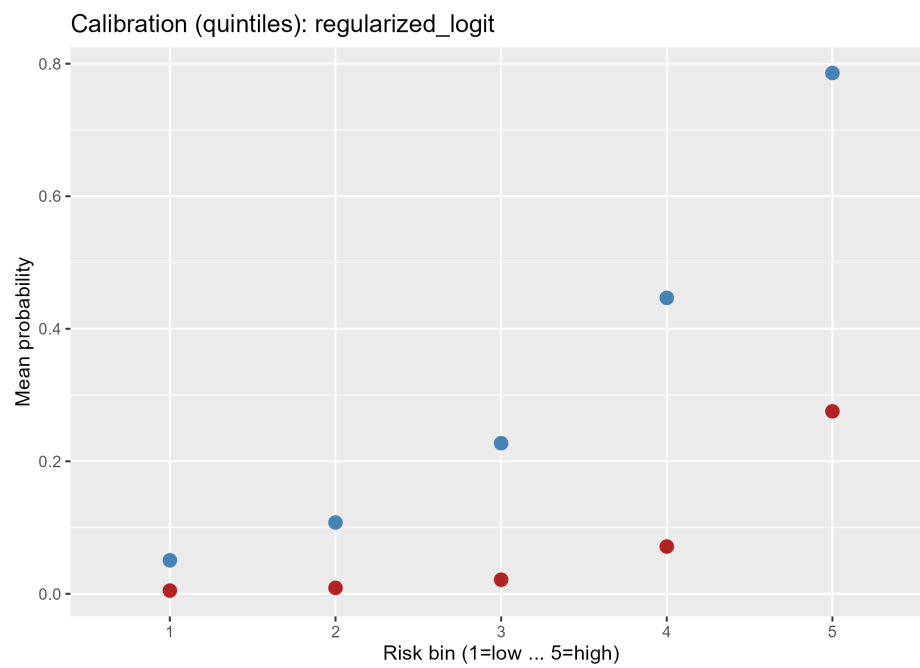


Figure 12: Regularized Logit: Calibration by Risk Quintile